

Nonnegative and positive factorizations

Exact nonnegative rank of regular polygon slack matrices

Difficulty: challenging **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: Challenging because matching lower bounds must hold for every polygon size despite an explicit upper construction; community importance connects structured NMF with optimal linear extended formulations.

Area: structured nonnegative matrix factorization

Context and notation

All factorizations are over the real numbers. For $X \in \mathbb{R}_{\geq 0}^{m \times n}$, define

$$\text{rank}_+(X) = \min\{r \geq 0 : X = WH, \quad W \in \mathbb{R}_{\geq 0}^{m \times r}, H \in \mathbb{R}_{\geq 0}^{r \times n}\}.$$

Problem statement

For each integer $n \geq 3$, define the $n \times n$ nonnegative matrix

$$S_n(i, j) = \cos(\pi/n) - \cos((2i + 1 - 2j)\pi/n), \quad 0 \leq i, j < n.$$

It is the slack matrix obtained from the vertices of the regular n -gon on the unit circle and its supporting facet inequalities.

Question

With $k = \lceil \log_2 n \rceil$, is it true for every $n \geq 3$ that

$$\text{rank}_+(S_n) = \begin{cases} 2k - 1, & 2^{k-1} < n \leq 2^{k-1} + 2^{k-2}, \\ 2k, & 2^{k-1} + 2^{k-2} < n \leq 2^k? \end{cases}$$

References

Arnaud Vandaele, Nicolas Gillis, François Glineur, and Daniel Tuytens, *Heuristics for Exact Nonnegative Matrix Factorization*, Journal of Global Optimization **65** (2016), 369–400, §6.2, Conjecture 2. Vandaele, Gillis, and Glineur, *On the Linear Extension Complexity of Regular n -gons*, Linear Algebra and its Applications **521** (2017), 217–239, proves the corresponding upper bound. Nicolas Gillis, *Nonnegative Matrix Factorization*, SIAM, 2020, §3.6.3.4, p. 90.

Status check — 2026-09-10

Rechecked Baeckelant–Vandaele–Gillis v2, Appendix A.2, Table 7, and searched for later regular-polygon rank resolutions. The conjectured value is attained and proved for substantive ranges, including n from 5 through 16, but gaps remain, for example at $n=17$. The upper formula remains conjecturally sharp for the whole family; no full resolution was located.

Reviewed partial result — 2026-09-13

Sidney Holden (Center for Computational Biology, Flatiron Institute, Simons Foundation) proves in [Theorem 1.1](#) that

$$\text{rank}_+(S_n) = 9, \quad n = 17, 18, 19, 20.$$

The computer-assisted lower bound covers arbitrary real factorizations, including nonsimple lifts and nongeneric projections; exact nine-term factorizations give the matching upper bounds. Sections 2–7 supply the rank-balance reduction, finite shadow cover, degeneration argument and characteristic-zero justification of the modular rank certificates. The argument also gives the lower bound nine for every size at least 21 by the eight-facet vertex bound in Section 2; together with the known upper construction this settles sizes 21–24.

The [independent informal Codex AI-agent audit](#) checks the mathematical reductions and reproduces the complete computational pipeline. This is partial affirmative progress: the original equality for every size remains open, beginning with $9 \leq \text{rank}_+(S_{25}) \leq 10$. In particular, sizes 25–30 and 33–42 remain unresolved by this submission; no general formula for the remaining sizes is proved. **NR-01 remains Partially resolved.** No Lean verification or external human peer review is claimed.

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